

Himesh Rathna Duddala

Fairfax, VA (DC Metro) · hduddala@gmu.edu · 240-797-7519 · [linkedin.com/in/hduddala](#) · [github.com/hduddala](#)

Software and AI/ML engineer who ships **production systems end-to-end**, not demos. Over the past year: multi-tenant LLM pipelines with prompt-injection guarding, LangGraph multi-agent systems holding **99.5% uptime** in low-connectivity field environments, and graph-neural-network research whose outputs are used by **NOAA / VIMS** teams — built with the reliability scaffolding (idempotency, quality gates, integrity tests) most people skip.

EXPERIENCE

Software Engineer

SOLE DEVELOPER

Inroads LLC — Content Rush

Jun 2026 – Present

Frederick, MD

- Sole engineer building an AI-powered social-media distribution platform on **Laravel + Vue.js**, owning architecture through deployment.
- Multi-tenant data isolation via Eloquent global scopes, enforced by scoping tests; AI content-drafting service gated by a CI quality check (structure, banned-word, prompt-injection, and LLM-judged brand-voice scoring).
- OAuth multi-platform publishing (LinkedIn, Facebook, Instagram, X, TikTok) with encrypted token storage; scheduled publishing + analytics on **Laravel Horizon / Redis** with idempotency and overdue-post detection.
- Resumable, chunked, direct-to-object-storage upload pipeline for large video.

Oceanic AI Research Fellow

Jan 2026 – May 2026

Williamsburg, VA

- William & Mary — Commonwealth Cyber Initiative (CCI)
- Computer-vision pipeline for marine imagery combining **YOLOv8** detection with graph neural networks; outputs used by VIMS / NOAA teams.
 - Adversarial-robustness work hardening models against perturbed inputs; GNN biomass modeling trained on **A100 GPUs via SLURM / HPC**.

Agentic AI Intern

DIGITAL TWIN SYSTEMS

Kashmir World Foundation

Jan 2026 – Jun 2026

Fairfax, VA

- Built **LangGraph** multi-agent workflows for a 50-node LoRa mesh network — automated sensor diagnostics and anomaly detection at 99.5% uptime in field conditions.
- Fault-tolerant RAG deployed across **AWS / Azure** for field resilience.

Full-Stack Software Engineering Intern

May 2024 – Aug 2024

Rise Consultancy

Vijayawada, India

- Shipped features and REST APIs for an admissions platform with JWT auth + role-based access control (RBAC).
- Containerized deploys via **Docker + GitLab CI/CD to AWS**.

Learning Assistant & Undergraduate Teaching Assistant

Aug 2023 – May 2026

George Mason University

Fairfax, VA

- Mentored students in Python and data structures through 1:1 tutoring and group workshops; coached debugging to build independent problem-solving.

PROJECTS

Replic-AI / BrandPilot

HACKPRINCETON 2025

Next.js · FastAPI · Supabase · Redis

- AI social-media agent that monitors X and Reddit for brand mentions, auto-drafts on-brand replies and daily posts, and routes chat-native approvals to founders via iMessage.

Real-Time Conversational Fact Checker

Python · Hugging Face · NER

- Live fact-checking chatbot for Slack/Discord; detects and classifies claims with Transformers and verifies against Google Knowledge Graph + Wikipedia in **under 2 seconds**.

TECHNICAL SKILLS

LANGUAGES	Python, Java, C, C++, JavaScript / TypeScript, SQL, Bash, R, PHP	AI / ML	PyTorch, TensorFlow, Hugging Face, LangChain, LangGraph, scikit-learn, YOLOv8, GNNs, XGBoost, RAG, multi-agent orchestration, adversarial training
WEB	React, Next.js, Node / Express, FastAPI, Flask, Spring Boot, Laravel, Vue.js	DATA / INFRA	PostgreSQL, Supabase, Redis, Laravel Horizon, vector / RAG retrieval
CLOUD / DEVOPS	AWS (EC2, RDS, S3), Azure, Docker, HPC / SLURM (A100), GitLab CI, GitHub Actions	PRACTICES	Multi-tenant architecture, OAuth / JWT + RBAC, event / queue pipelines, idempotency & exactly-once delivery, prompt-injection guarding, LLM-as-judge quality gates

EDUCATION

B.S. Computer Science — George Mason University

May 2026 · GPA 3.6 / 4.0

Coursework: Machine Learning, Natural Language Processing, Data Structures, Algorithms, Operating Systems, Secure Programming

Authorized to work in the U.S. - F-1 student on STEM OPT